



Defensive Security Project

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Monitoring Environment

Scenario

- We are a SOC Team for VSI - Virtual Space Industries
- Rumors of a cyberattack from competitor JobeCorp
- 2 sets of logs
- Analysis and comparison using Splunk and an Add-on



The background of the slide is a dark, abstract geometric pattern. It consists of numerous triangles of varying sizes and shades of dark red and black, arranged in a complex, non-repeating pattern that creates a sense of depth and movement. The triangles are oriented in different directions, some pointing up, some down, and some towards the center, creating a dynamic visual effect.

Detection Backfill

Detection Backfill

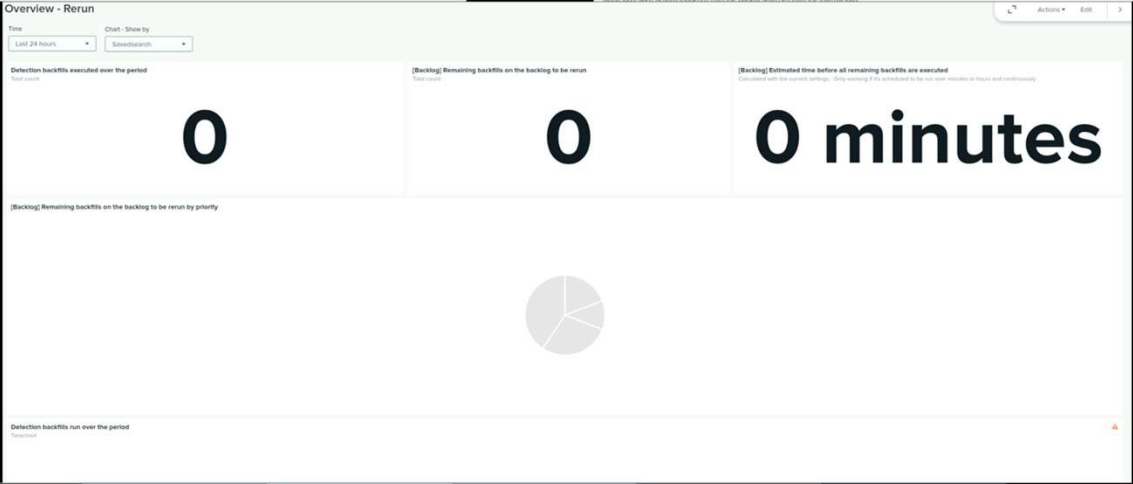
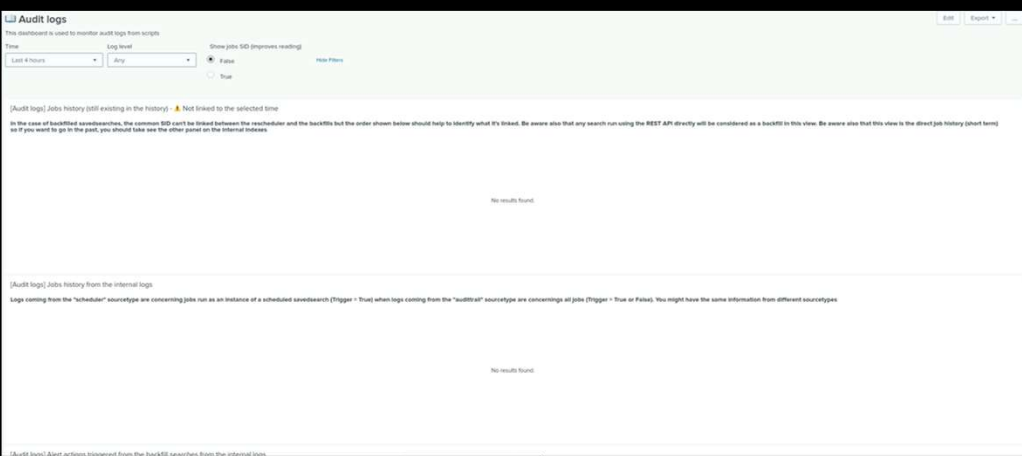
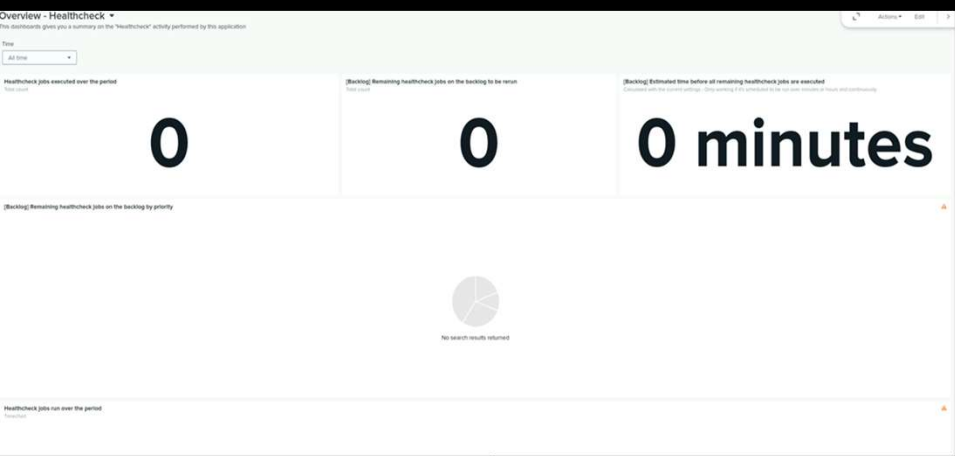
Detection Backfill is a logging system and a health scanner that takes live logs from the servers to be scanned and used for later.

- The health scanner schedules scans on the logs to find any issues with the server with irregularities and possible attacks.
- The addon can also start a scan after a server blackout cause by stupid people or natural causes.
- Logs made by detection backfill could also be used as to research baselines and/or to add it to AI learning for perfecting AI tools.

Detection Backfill

- Recovery of data.
- Health hazards in the logs would be detected and logged.
- Automatic scanning start post blackout.
- Good for tracking baselines.
- Helpful for feeding data to AI for training.
- Easy gathering of logs for archiving.
- Old logs are helpful to anyone needing to see if the company is operating within the scope of the companies operations.

Detection Backfill



Windows Logs

Logs Analyzed

1

Windows Logs

- signature
- signature_id
- user
- status
- severity

signature_id

15 Values, 100% of events

Selected

Reports

[Average over time](#) [Maximum value over time](#) [Minimum value over time](#)

[Top values](#) [Top values by time](#) [Rare values](#)

[Events with this field](#)

Avg: 4475.4013434089 **Min:** 1102 **Max:** 4743 **Std Dev:** 879.9767461845869

Top 10 Values	Count	%
4672	342	7.179%
4743	340	7.137%
4648	337	7.074%
4739	329	6.906%
4624	323	6.78%
4718	321	6.738%
4726	318	6.675%
4673	317	6.654%
4720	313	6.57%
4689	309	6.486%

Reports—Windows

Designed the following reports:

Report Name	Report Description
signature and signature_id report	provides a table of signatures and signature IDs, duplicates removed
Severity Levels of Windows Logs	shows severity level counts, including percentage
Success vs Failure of Windows Activities	provides a comparison of success / failure of Windows activities

Images of Reports—Windows

source="windows_server_logs.csv" host="Windows_server_logs" sourcetype="csv" | stats count by signature, signature_id | sort signature

4,764 events (before 11/26/24 2:03:06.000 AM) No Event Sampling

Events (4,764) Patterns Statistics (1) Visualization

20 Per Page / Format Preview

signature	signature_id	count
1 computer account was deleted	6743	148
1 login was attempted using explicit credentials	6646	137
1 privileged service was called	6673	117
1 process has exited	6693	101
1 user account was changed	4718	299
1 user account was created	4718	213
1 user account was deleted	4718	218
1 user account was locked out	4718	109
10 account was successfully logged on	6514	123

New Search

source="windows_server_logs.csv" host="Windows_server_logs" sourcetype="csv" | stats count by severity | eventstats sum(count) as Total | eval Percentage = round((count / Total) * 100, 2) | sort -count

4,764 events (before 11/26/24 2:16:42.000 AM) No Event Sampling

Events (4,764) Patterns Statistics (2) Visualization

20 Per Page / Format Preview

severity	count	Percentage	Total
Informational	4435	93.09	4764
high	329	6.91	4764

New Search

source="windows_server_logs.csv" host="Windows_server_logs" sourcetype="csv" | stats count by status | sort -count

4,764 events (before 11/26/24 2:26:25.000 AM) No Event Sampling

Events (4,764) Patterns Statistics (2) Visualization

20 Per Page / Format Preview

status	count
success	4622
failure	142

Alerts—Windows

Designed the following alerts:

Alert Name	Alert Description	Alert Baseline	Alert Threshold
Failed Windows Activity	Hourly threshold for failed Windows activity	25 per hour	30

JUSTIFICATION: The results for the Failed Windows Activity alert on March 24, 2020 ranged from 4 to 20 per hour. The baseline was set to 25 based on a standard deviation calculation (~4) to account for additional events that could occur on a normal day, but did not on March 24, 2020. The threshold was set to 30 to prevent false positives.

Alerts—Windows

Designed the following alerts:

Alert Name	Alert Description	Alert Baseline	Alert Threshold
Account Successfully Logged On	The number of successful logons per hour has exceeded the threshold.	50	55

JUSTIFICATION: The results for the Account Successfully Logged On alert on March 24, 2020 ranged from 16 to 42 per hour. The baseline was set to 50 based on a standard deviation calculation (~6) to account for additional events that could occur on a normal day, but did not on March 24, 2020. The threshold was set to 55 to prevent false positives.

Alerts—Windows

Designed the following alerts:

Alert Name	Alert Description	Alert Baseline	Alert Threshold
A User Account Was Deleted	The number of user accounts deleted per hour has exceed the threshold	55	60

JUSTIFICATION: The results for the A User Account Was Deleted alert on March 24, 2020 ranged from 14 to 44 per hour. The baseline was set to 55 based on a standard deviation calculation (~9) to account for additional events that could occur on a normal day, but did not on March 24, 2020. The threshold was set to 60 to prevent false positives.

Dashboards—Windows



Apache Logs

Reports—Apache

Designed the following reports:

Report Name	Report Description
HTTP Post Method Alert	Shows the types of request being made
Top Referring Domains	Looks for suspicious referrals
HTTP Response Code Summary	Provides a way to check for any suspicious levels of HTTP responses

Images of Reports—Apache

HTTP Methods Activity Edit More Info Add to Dashboard

All time Export

✓ 10,000 events (before 11/26/24 2:02:08.000 AM)

4 results 20 per page

method	count
GET	9851
HEAD	42
OPTIONS	1
POST	106

Top Referring Domains Edit More Info Add to Dashboard

All time Job Filter Refresh Download

✓ 10,000 events (before 11/26/24 2:03:28.000 AM)

10 results 20 per page

referrer_domain	count	percent
http://www.sensiocomplete.com	3838	61.25888
http://testcomplete.com	3001	33.768755
http://www.google.com	123	2.875349
https://www.google.com	105	1.771354
http://stackoverflow.com	34	0.575946
http://www.google.fr	31	0.523838
http://r-chassis.co.nz	29	0.483386
http://logstash.net	28	0.472814
http://www.google.es	25	0.427795
https://www.google.co.uk	23	0.388855

HTTP Response Code Summary Edit More Info Add to Dashboard

All time Job Filter Refresh Download

✓ 10,000 events (before 11/26/24 2:09:33.000 AM)

8 results 20 per page

status	count
200	91246
206	45
301	1614
304	445
403	2
404	213
416	2
500	3

Alerts—Apache

Designed the following alerts:

Alert Name	Alert Description	Alert Baseline	Alert Threshold
Non-US Traffic Alert	Checks to see how much search traffic is from outside the United States in a given hour	the average is less than 100 visit in an hour	the alert is triggered if the amount exceed 125 in an hour

JUSTIFICATION: while traffic from other countries is normal there is not a lot in an hours time so any large amount in a short time could be an issue

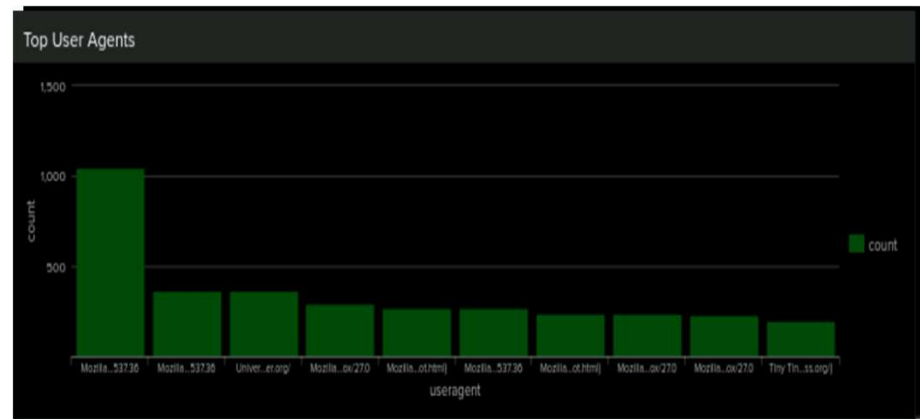
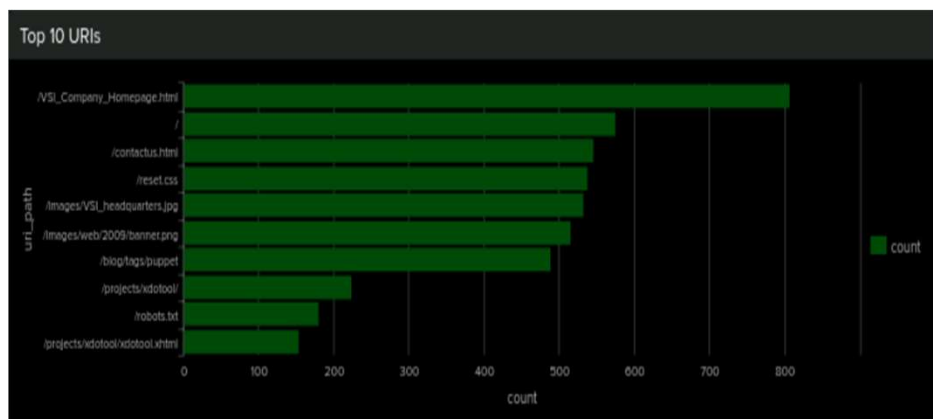
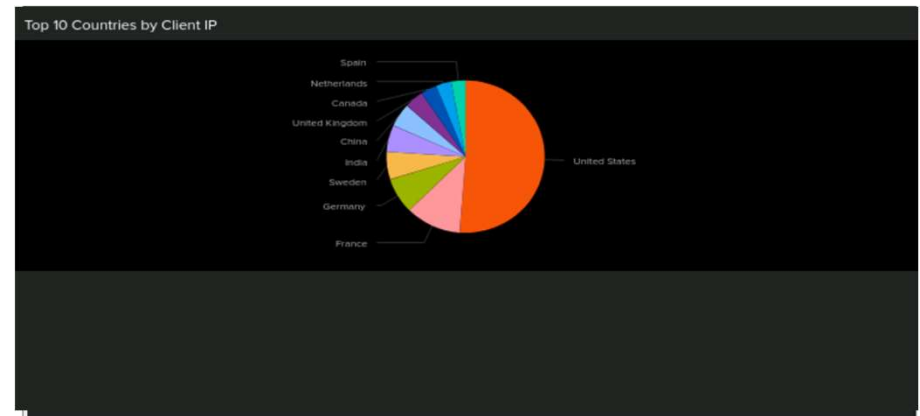
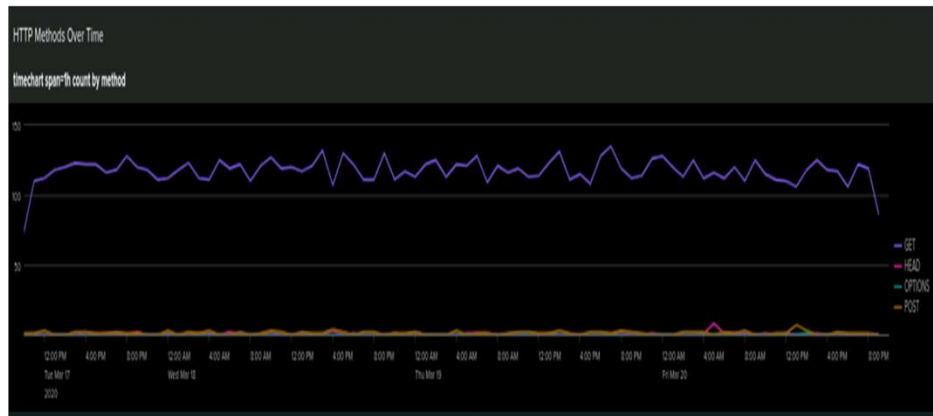
Alerts—Apache

Designed the following alerts:

Alert Name	Alert Description	Alert Baseline	Alert Threshold
HTTP Post Method	Checks the number of posts to the website	3 is the average number of posts	Anything more than 6 in an hour

JUSTIFICATION: The average number of post is around 3 however if there is a increase the number who have to be larger than 6

Dashboards—Apache



The background of the slide is a dark, abstract geometric pattern. It consists of numerous triangles of varying sizes and shades of dark red and black, creating a complex, low-poly aesthetic. The triangles are arranged in a way that they seem to recede or project, giving a three-dimensional feel. The overall color palette is very dark, with the red being a deep, almost blackish-red.

Attack Analysis

Attack Summary — Reports — Windows

Summarize your findings from your **reports** when analyzing the attack logs.

- An aberration in activity logs (signature & signature_id) indicates an attempted **Brute Force** attack upon the VSI Windows Server.
- On the day of the attack, there was a 26.7% increase in successful activities.
- The number of “High” severity counts increased from 6.9% to 20% on the day of the attack.

Attack Summary – Alerts – Windows

Summarize your findings from your **alerts** when analyzing the attack logs. Were the thresholds correct?

- Two alerts would have been triggered (Failed Windows Activity and Account Successfully Logged On) - these appear to be correct - even compensating for additional events that may not have been represented in one 24 hour period, the alerts were triggered
- One would not have been triggered (A User Account Was Deleted) - also appears to be correct - event counts per hour on attack day was actually lower than the original counts

Attack Summary—Windows

Summarize your findings from your dashboards when analyzing the attack logs.

- There was an increase in attempts to reset passwords and account lockouts from 12am - 3am and 8am - 11am. These times correlate with an increase in count of user_a and user_k. This was a bruteforce attack → user_a attacked from 12am- 3am and primarily locked out accounts, user_k attacked from 8am - 11am and primarily attempted to reset passwords

Screenshots of Attack Logs



Attack Summary—Apache

Summarize your findings from your reports when analyzing the attack logs.

- Between 5pm and 9pm there appeared to be a brute force attack where it appears that someone attempted to login into our system during this timeframe

Attack Summary—Apache

Summarize your findings from your alerts when analyzing the attack logs. Were the thresholds correct?

- From the logs that were monitored, the thresholds for alerts appears to be set to the correct levels. The alert would activate for Ukraine

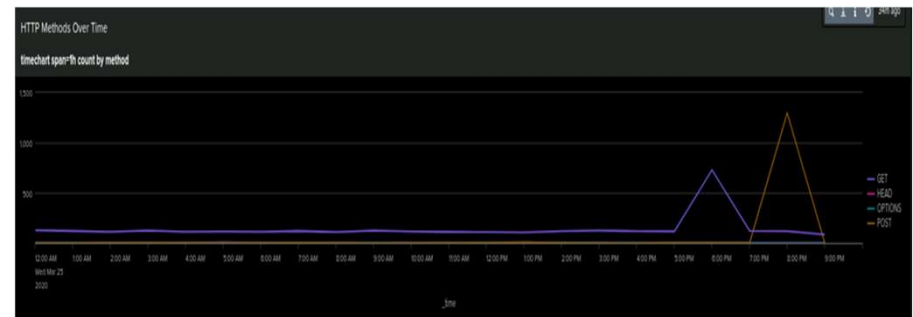
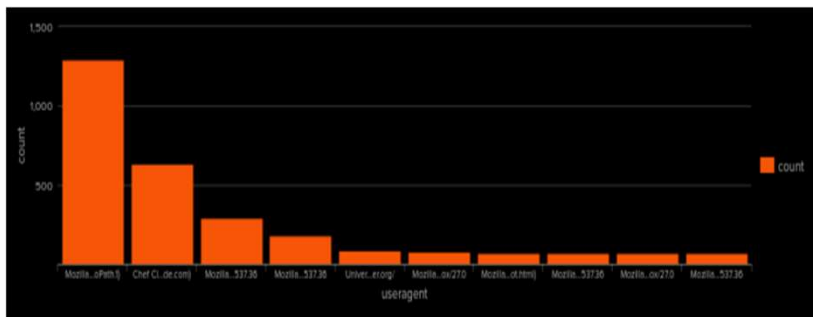
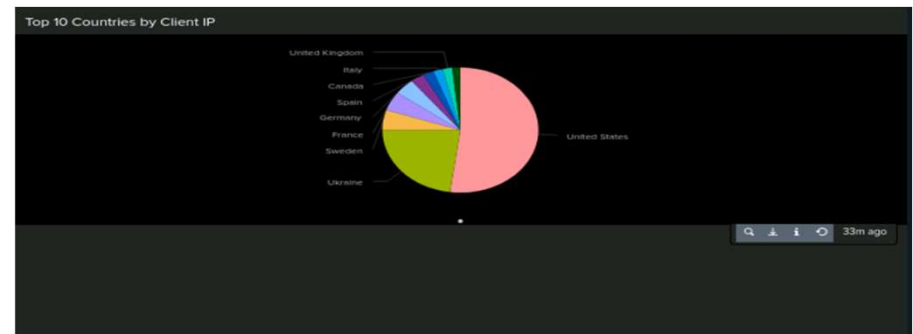
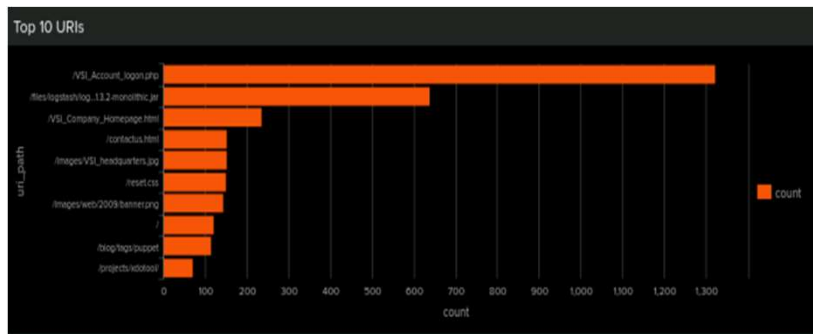
Attack Summary—Apache

Summarize your findings from your dashboards when analyzing the attack logs.

- An Attacker based in the Ukraine, using the Mozilla user agent attempted a brute force attack through the VSI account logon. this appeared to be an attempt to gain access to the server to gain information.

Screenshots of Attack Logs

[Optional: Place images that illustrate attack findings here.]





Summary and Future Mitigations

Project 3 Summary

- What were your overall findings from the attack that took place?
 - Brute force attacks were made against the Windows and Apache servers.
 - Windows server - 12am-3am, and 8am-11am
 - Apache server 5pm - 9pm
- To protect VSI from future attacks, what future mitigations would you recommend?
 - For the Windows server:
 - create a robust password policy with MFA for users
 - limit the number of login attempts
 - For the Apache server:
 - improve the VSI intrusion protection software/platform